ALGB-01 • SIDE A (PAGES 01–04) • BENCH FOLD: (1) FOLD BOTTOM HALF BEHIND TOP, (2) FOLD RIGHT HALF BEHIND LEFT SO PAGE 01 IS ON TOP

◀ FOLD 2: FOLD RIGHT BEHIND LEFT (01 ON COVER) | UNFOLD STEP 1: OPEN RIGHT ▶

ALGEBRA FACTORING STUDIO & POLYNOMIAL TILES

Turn Quadratics Into Rectangles: +x², +x, and +1 Area Tiles

01 02 03 04

SKU: ALGB-01 • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

WHAT IS THIS BOX FOR? WHY "X-SQUARED" IS LITERALLY A SQUARE.

Snap the corner rails together, slide in your tiles, and turn scary quadratic equations into a satisfying tabletop jigsaw puzzle!

THE AREA MODEL: MULTIPLYING BY BUILDING RECTANGLES

The area of any rectangle is Width × Height. Our tiles turn algebra expressions into physical shapes you can arrange on your desk:

- **Quadratic Tile (x²): A big x-by-x square (38.6 × 38.6 mm)**
- **Linear Rod (x): An x-by-1 rectangle (38.6 × 19.0 mm)**
- **Unit Square (1): A 1-by-1 little square (19.0 × 19.0 mm)**

Green = Positive (+x², +x) | Purple = Negative (−x², −x) | Yellow/Red = (+1, −1).

CORNER L-RAIL FACTORING FRAME

Snap the two L-rails together on your desk. Arrange your tiles inside the corner into a solid rectangle – then read the factors along the outer edges!

WHAT'S INSIDE YOUR BOX (34 PIECES)

- 30 Algebra Tiles: 2× x² squares, 10× x rods, 18× unit squares (+ and −)
- 2× Interlocking Corner L-Rails + 1× Storage Box & Dual-Stage Lid
- Keep this folded guide inside your box lid for examples & part IDs!

★ YOUR FIRST 60 SECONDS WITH THIS BOX

START HERE!

1. TEST WHY [+x] IS INCOMMENSURABLE:
Line up Yellow [+1] tiles (19.0mm) along a Green [+x] rod (38.6mm): x is a true variable length!

2. BUILD YOUR FIRST RECTANGLE:
Snap the two Gray L-Rails together and pack 1×[+x²], 5×[+x], and 6×[+1] into a (x+2)(x+3) rectangle.

3. PRESS THE BOX ICON KEY [↱][POLY]:
Press [↱] [POLY] on your SC-32 and enter 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] to find roots −2 and −3!

EQN

BOX ICON = CALCULATOR [↱][EQN] & [↱][POLY] QUADRATIC SOLVER

The button icon embossed on your box lid (left) is the exact calculator key ([↱][EQN] & [↱][POLY])! Build every lesson with your blocks, then press the key.

L-RAIL FACTORING FRAME ↔ BOX [↱][POLY] SOLVER ICON

STEP A: HANDS-ON WITH TILES IN L-RAILS

1:1 CAD SCENE

Top Rail	Left Rail	Box Icon Key
Reads Width (x + 3)	Reads Height (x + 2)	[↱][POLY] Roots −2, −3

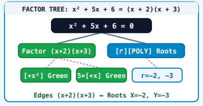
▲ **Snap the two Gray L-Rails together at 90° and pack your tiles** inside–then verify roots with [↱][POLY]!

EQN

STEP B: TRY IT ON YOUR CALCULATOR

BOX ICON: [↱][EQN] & [↱][POL

FACTOR TREE: x² + 5x + 6 = (x + 2)(x + 3)



Edges (x+2)(x+3) ↔ Roots X=−2, Y=−3

PRESS BOX ICON & RPN KEYS (SC32):

1. [↱][POLY]:
↱ POLY 1 A 5 B 6 C

2. SOLVE ROOTS:
0 D EXEC X=−2, Y=−3

✓ DISPLAY (X) = −2

KEYSTROKES: [↱] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] 3
Press [↱][POLY] (the equation icon on your box!) to verify roots −2 and −3 of (x+2)(x+3)!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

HOW IT WORKS: THE CORE IDEA

Area = Length × Width: Factoring Is Just Building a Rectangle

01 02 03 04

SKU: ALGB-01 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

POLYNOMIALS AS GEOMETRIC PUZZLES

When you multiply two binomials on paper (FOIL), you get four parts:
(x + a)(x + b) = x² + ax + bx + ab

Our Algebra Tiles turn those four parts into a literal rectangle: every piece has a real physical area:

- **Area of x² square = x × x**
- **Area of x rod = x × 1**
- **Area of unit tile = 1 × 1**

Factoring is just playing in reverse: grab the tiles for an expression, pack them into a solid rectangle, and read the side lengths!

WHY THE 'X' ROD HAS SMOOTH UNMARKED SIDES

Because x is a variable that can stand for any number, the x rod is smooth with no unit tick marks on its face – reminding your brain that x is an unknown variable length, while the 1-tiles are fixed constants.

THE ZERO-PAIR SUPERPOWER (+x AND −x = 0)

Place any positive tile next to its negative twin: (+x) + (−x) = 0, and (+1) + (−1) = 0! Adding a zero-pair doesn't change the total value, but it gives you the missing pieces needed to complete a rectangle.

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE

VISUAL PROOF

✗ **TRAP 1 – Writing (x + 3)² = x² + 9:**
✓ **FIX:** Build an (x+3)×(x+3) square in the rails–you immediately see the six Green [+x] wing rods you forgot!

✗ **TRAP 2 – Combining x² and x into '2x³':**
✓ **FIX:** A 38.6×38.6mm [+x²] square and a 38.6×19.0mm [+x] rod have different dimensions and cannot merge.

✗ **TRAP 3 – Guessing signs when factoring:**
✓ **FIX:** Purple/Red negative tiles (−x, −1) and Green/Yellow positive tiles (+x, +1) make zero-pairs visible.

CALCULATOR CHECK: [↱] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] → X = −2

TILE DIMENSIONS & COLOR GUIDE

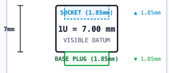
COMPONENT	DIMENSIONS	VALUE	COLOR
Quadratic (+x²)	38.6 × 38.6 mm	+x²	Green
Quadratic (−x²)	38.6 × 38.6 mm	−x²	Purple
Linear Rod (+x)	38.6 × 19.0 mm	+x	Green
Linear Rod (−x)	38.6 × 19.0 mm	−x	Purple
Unit Block (+1)	19.0 × 19.0 mm	+1	Yellow
Unit Block (−1)	19.0 × 19.0 mm	−1	Red
Corner L-Rails	123.0 × 19.9 mm	Guides	Gray
3x1 Storage Bin	126 × 42 × 49 mm	3x1x7U	Black

WHY YOUR BRAIN LOVES THIS

- Self-Checking Geometry: If your tiles form a solid rectangle with no missing corners or leftover pieces, your factoring is 100% correct!
- Custom Recessed Bays: Every tile has a home so cleanup takes 15 seconds.

1:1 CAD REFERENCE VIEW

1:1 BLOCK CAD ANATOMY



✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely:
Only 7.00mm true heights add!

BOX ICON & FORMULA QUICK-REFERENCE

Box Button Icon Key:
[↱][EQN] & [↱][POLY] Polynomial Solver

Rectangle Area Law:
Area = Width × Height = ax² + bx + c

Zero-Pair Identity:
[+x] + [−x] = 0 and [+1] + [−1] = 0

Complete the Square:
Split b rods in half → add (b/2)² units

SC-32 Quadratic Roots:
[↱][POLY] a[A] b[B] c[C] 0[D] [EXEC]

GEOMETRIC ALGEBRA • EVERY TRINOMIAL IS A RECTANGLE PUZZLE

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

EXAMPLE 1: HANDS-ON WALKTHROUGH

Build (x + 2)(x + 3) = x² + 5x + 6 Inside the Corner Rails

01 02 03 04

SKU: ALGB-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

1 DEPLOY RAILS & MODEL TRINOMIALS

GOAL: Factor the quadratic expression x² + 5x + 6

1. DEPLOY THE FACTORING CORNER RAILS:
Lock the two L-rails together on your desk. The inside corner establishes the Cartesian origin (0, 0).

2. GATHER THE POLYNOMIAL TILES:
Take one +x² tile, five +x linear rods, and six +1 unit squares.

3. ARRANGE INTO A SOLID RECTANGLE:
- Seat the +x² tile flush against the corner origin.
- Place three +x rods horizontally along the bottom rail.
- Place two +x rods vertically along the side rail.
- Fill the remaining corner void with the six +1 unit tiles!

READ FACTORS DIRECTLY FROM OUTER EDGES

Look at the outer dimensions of your completed rectangle:

- **Horizontal Width: One x length + three unit widths = (x + 3)**
- **Vertical Height: One x length + two unit heights = (x + 2)**

✓ **Physical Factoring: x² + 5x + 6 = (x + 2)(x + 3)**

EQN

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR

[↱][EQN] & [↱][POL

STEP 1 (BUILD RECTANGLE):
Pack 1×[+x²], 5×[+x], and 6×[+1] inside the Gray L-Rails to form edges (x + 2) and (x + 3).

STEP 2 (EVALUATE AT x=4):
Press 4 [ENTER] 2 [+] (= 6) then 4 [ENTER] 3 [+] (= 7) and [×] → X = 42 (matches 4²+5(4)+6!).

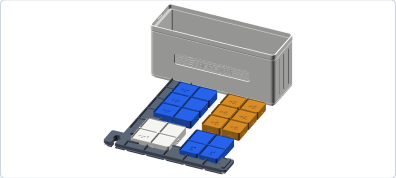
STEP 3 (SOLVE WITH [↱][POLY]):
Press [↱][POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] 3 → Y = −3, X = −2!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 42 → DISPLAY (X) = 42

LESSON 1: (x+2)(x+3) RECTANGLE ↔ RPN AREA EVALUATION

STEP A: HANDS-ON WITH TILES IN L-RAILS

1:1 CAD SCENE



Corner [+x²] Green	Five [+x] Green	Six [+1] Yellow
38.6 × 38.6 mm	38.6 × 19.0 mm	19.0 × 19.0 mm

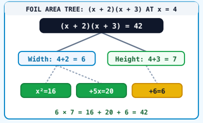
▲ **Slide +x², five +x rods, and six +1 units tightly into the** L-Rails: the outer edges read (x + 2) by (x + 3)!

EQN

STEP B: TRY IT ON YOUR CALCULATOR

BOX ICON: [↱][EQN] & [↱][POL

FOIL AREA TREE: (x + 2)(x + 3) AT x = 4



6 × 7 = 16 + 20 + 6 = 42

PRESS BOX ICON & RPN KEYS (SC32):

1. WIDTH x+2:
4 ENTER 2 + = 6

2. × HEIGHT:
4 ENTER 3 + × = 42

✓ DISPLAY (X) = 42

KEYSTROKES: 4 [ENTER] 2 [+] 4 [ENTER] 3 [+] [×]
Build the (x+2)(x+3) rectangle in the L-Rails, then test x=4 on the RPN stack: 6 × 7 = 42!
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 42 → X = 42

READ FACTORS ON THE RAILS • OUTER EDGES REVEAL (x + a)(x + b)

CONTINUE RIGHT → PAGE 04

EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

Zero-Pair Magic: Factoring x² − 4 (Difference of Squares)

01 02 03 04

SKU: ALGB-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

2 ZERO-PAIR ANNIHILATION: FACTOR x² − 4

THE PUZZLE: How do you build a rectangle with x² and −4?
If you place +x² and four −1 tiles, there is a giant corner gap.
You cannot form a rectangle without adding linear rods!

THE ZERO-PAIR BREAKTHROUGH:

- Add two positive +x rods AND two negative −x rods.

(+2x) + (−2x) = 0 (The total value of the expression is unchanged!)

- Place the two +x rods along the horizontal base.
- Place the two −x rods along the vertical side.
- Fill the remaining corner with four red −1 unit tiles!

(+2) × (−2) = −4 (Matches the constant term perfectly!)

✓ DIFFERENCE OF SQUARES PHYSICAL THEOREM

x² − a² = (x + a)(x − a)

x² − 4 = (x + 2)(x − 2)

The outer rail lengths now read (x + 2) and (x − 2) directly.
The positive and negative linear terms cancel out in the middle!

AHA! DISCOVERY MOMENT

Students physically witness zero-pairs in action. Annihilation is not just an abstract symbol manipulation; it is a physical symmetry that completes geometric shapes!

★ 2-PLAYER GAME: 'SECRET RECTANGLE PUZZLE' (2 MINS)

Player A picks a factor pair (like (x+3)(x−1)), expands it, and hands the simplified pile of tiles (1×[+x²], 2×[+x], 3×[−1]) to Player B. Player B must add zero-pairs ([+x]+[−x]) until all tiles **pack into a tight rectangle inside the L-Rails, then verify roots with [↱][POLY]!**

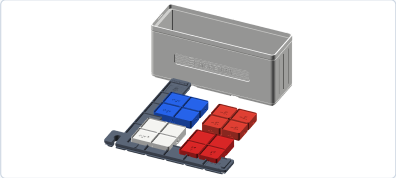
⚙ WHERE THIS SHOWS UP IN ARCHITECTURE, GRAPHICS & AI

Every time al-Khwarizmi solved an equation in 820 AD, he literally 'completed the square' by moving rectangular strips around a square! Today, parabolic trajectories in physics and polynomial **splines in 3D animation use this exact root-and-area geometry.**

LESSON 2: DIFFERENCE OF SQUARES (x²−4) ↔ [↱][POLY]

STEP A: HANDS-ON WITH TILES IN L-RAILS

1:1 CAD SCENE



Top-Left: [+x²]	Wings: +2x & −2x	Corner: 4×[−1]
38.6 × 38.6 mm	Zero-Pair Cancel	x² − 4 = (x+2)(x−2)

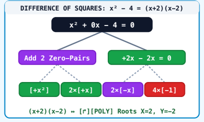
▲ **Adding two (+x, −x) zero-pairs completes the (x+2)(x−2)** rectangle and reveals roots x = +2 and x = −2!

EQN

STEP B: TRY IT ON YOUR CALCULATOR

BOX ICON: [↱][EQN] & [↱][POL

DIFFERENCE OF SQUARES: x² − 4 = (x+2)(x−2)



(x+2)(x−2) ↔ [↱][POLY] Roots X=2, Y=−2

PRESS BOX ICON & RPN KEYS (SC32):

1. x²−4 POLY:
↱ POLY 1 A 0 B 4 +/− C

2. READ ROOTS:
0 D EXEC X=2, Y=−2

✓ DISPLAY (X) = 2

KEYSTROKES: [↱] [POLY] 1 [A] 0 [B] 4 [+/−] [C] 0 [D] [EXEC] [RCL] 1 [R
Adding two [+x]/[−x] zero-pairs completes (x+2)(x−2) and matches [↱][POLY] roots X=2, Y=−2!
✓ 4-REGISTER STACK: T: −2 | Z: 0 | Y: −2 | X: 2 → X = 2

ZERO-PAIR SYMMETRY • WHY (+x) AND (−x) CANCEL IN MID-AIR

STEP 3: FLIP SHEET OVER ↻ PAGE 05

◀ PAGE 03 (LEFT) → PAGE 04 (RIGHT) • THEN FLIP SHEET OVER FOR PAGES 05–08 ▶

◀ PAGE 05 (CHALLENGES) → PAGE 06 (HOW TO STACK BLOCKS BACK IN) ▶

EXAMPLE 3 & DESK CHALLENGES

Factoring Trinomials: $x^2 + 5x + 6$ + 3 Desk Challenges

③ FACTORING TRINOMIALS: $x^2 + 5x + 6 = (x + 2)(x + 3)$

THE GOAL: Factor $x^2 + 5x + 6$ into a solid rectangle using the corner rails!

1. SEAT [+x²] AT THE ORIGIN:
Place the large green quadratic square tile at the intersection of the two L-rails.

2. SPLIT FIVE [+x] RODS (2 & 3):
Place two green linear rods vertically and three green rods horizontally along rails.

3. FILL SIX [+1] UNITS:
Drop six yellow constant squares into the 2 × 3 corner to complete the rectangle.

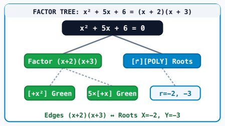
✓ READ BINOMIAL FACTORS DIRECTLY OFF THE RAILS

The outer horizontal rail length is $(x + 3)$; the vertical rail length is $(x + 2)$.
Area = Length × Width: $x^2 + 5x + 6 = (x + 2)(x + 3)$ with zero pencil guessing!
Roots are where factors equal zero: $x + 2 = 0 \rightarrow x = -2$, and $x + 3 = 0 \rightarrow x = -3$.
Verify on calculator: [↱][POLY] 1 [A] 5 [B] 6 [C] [EXEC] finds roots x = −2, −3!

PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE



STEP B: PRESS BOX ICON & RPN KEYS (SC32)

TACTILE KEYSTROKE WALKTHROUGH:

1. [↱][POLY]:
↱ POLY 1 A 5 B 6 C

2. SOLVE ROOTS:
0 D EXEC X=−2, Y=−3

✓ CALCULATOR DISPLAY (X) = −2

EQN FULL KEYSTROKES: [↱] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] 3
✓ 4-REGISTER STACK: T: −3 | Z: 0 | Y: −3 | X: −2 → X = −2

3 DESK CHALLENGES – TRY THESE NOW!

WARM-UP #1 **Arrange 1x [+x²], 4x [+x], and 3x [+1] into a rectangle.**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #1 below!

LEVEL-UP #2 **Use [+x] and [−x] zero-pairs to factor $x^2 - x - 2$.**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #2 below!

BRAIN-MELTER #3 **How many [+1] tiles complete the square for $x^2 + 4x$?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)
#1: $(x + 1)(x + 3) = x^2 + 4x + 3$ | #2: $(x - 2)(x + 1)$
#3: $(4/2)^2 = 4$ unit tiles → $(x + 2)^2$

FOR PARENTS & TEACHERS (Grades 7–11 (Pre-Algebra, Algebra I & Algebra II))

- CCSS.MATH.CONTENT.HSA.SSE.A.2: Use the structure of an expression to rewrite it.
- CCSS.MATH.CONTENT.HSA.REI.B.4.A: Use the method of completing the square.
- TEKS 111.39.c.10.E: Factor trinomials and difference of two squares.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

BOSS CHALLENGE #4 **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([↱][EQN] & [↱][POLY]) and keystrokes on your calculator:
- [↱] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] 3 → X = −2

CHECK OFF YOUR MASTERY MILESTONES:

☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
☐ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

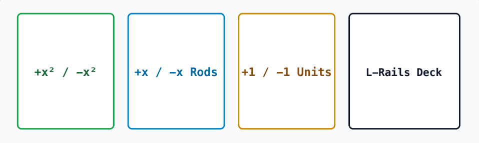
NEXT: HOW TO PACK UP → PAGE 06

HOW TO STACK IT ALL BACK IN

The 10-Second Self-Checking Pack-Up Puzzle

TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

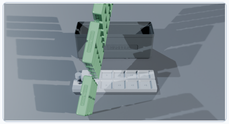
Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:



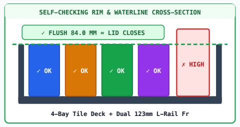
✓ CUSTOM-SHAPED BAYS FOR EVERY TILE TYPE

The box is divided into four dedicated bays: Quadratic Squares (+x²/−x²), Linear Rods (+x/−x), Unit Constants (+1/−1), and the two Corner L-Rails. When all 30 tiles and 2 rails are stacked in their matching bays, the top surface sits completely flat so the lid clicks shut with zero rattle.

1. STORAGE CASE & COMPARTMENTS



2. WHY THE LID SELF-CHECKS EVERY PIECE


SELF-CHECKING RIM & WATERLINE CROSS-SECTION
✓ FLUSH 84.0 MM = LID CLOSES
✓ OK
✓ OK
✓ OK
✓ OK
✗ HIGH
4-Bay Tile Deck + Dual 123mm L-Rail Fr

▲ **SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:**
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

THE 10-SECOND PACK-UP ROUTINE

HOW TO STACK EVERYTHING BACK IN:

1. SORT BY COLOR & NUMBER:
Group matching blocks or tiles together on your desk first.

2. STACK & DROP INTO LABELED BAYS:
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!

3. TUCK THIS CARD IN THE LID & SNAP SHUT:
Fold this guide back to [01] and rest it right on top before closing.

CORNER L-RAILS & STAGE LID

- Place the two L-Rails on your desk to form a 90° factoring frame for any rectangle.
- Why doesn't [+x] equal a whole number of [+1] tiles? So you never confuse x with a fixed constant!
- Easy care: wipe clean with a damp cloth or lukewarm soapy water.

SHORT A PIECE AT CLEAN-UP TIME?

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

✓ ALGB-01 4-POINT CLEAN-UP CHECKLIST

1. SORT 4 TILE BAYS:	Stack +x²/−x² (38.6mm), +x/−x (38.6×19mm), and +1/−1 (19mm) in their bays.
2. SEAT L-RAILS:	Nest both Gray Corner L-Rails (123.0 mm) flat along the side rail deck.
3. FOLD CARD TO [01]:	Fold this reference booklet to Panel [01] and lay inside the lid.
4. SNAP LID FLUSH:	Close the 3×1 box lid—all 34 pieces lock with zero rattle!

MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)
Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2). All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

CUSTOM TILE BAYS • LID ONLY SLIDES SHUT WHEN ALL 34 PIECES ARE HOME

NEXT: CONNECTED LAB ▼ PAGE 07

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

THE STACKCALC CONNECTED LAB

How Your Kits & RPN Calculator Stack Work Together

ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

WHERE YOUR BOX FITS IN THE 9-KIT LAB

BASE-10 **Ten-Frames Lab** Early Counting & Regrouping	DOMI-01 **Math Domino Studio** Doubles & 12-Frame Part-Whole	FRAC-01 **Fraction Studio** Proportional Towers & 4-Way Faces
PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers	ALGB-01 ◀ YOUR BOX **Algebra Studio** Quadratic Area Arrays & Factoring	TIME-01 **Planetary Clock** 12:1 Geared Hands & Time Wedges
PROB-01 **Probability Dice** Fair Polyhedrals & 10-Sector Spinner	VOLV-01 **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals	SC-32-HW **SC-32 RPN Calculator** 4-Register Stack Scientific Engine

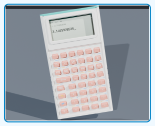
WHY THIS MATTERS: ZERO "THROW-AWAY" MATH

In most classrooms, every grade uses a totally unrelated plastic toy. In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. ALGB-01 BOX LID ICON

EQN
[↱][EQN] & [↱][POLY]
EMBOSSED ON BOX LID
Build First → Press Key!

2. SC-6 ELEMENTARY RPN



3. SC-32 SCIENTIFIC RPN

▲ **BOX ICON ↔ CALCULATOR KEY CONTINUITY:**
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

HOW BLOCKS CONNECT TO THE CALCULATOR

SHARED MODULAR FOOTPRINT & HEIGHTS

- Constant [+1] tiles (19.0 × 19.0 mm) match the universal StackCalc grid footprint.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

WHY WE CALL IT "STACKCALC" (THE RPN SECRET)

- Bridges physical factoring to the SC-32 calculator: test any root x by evaluating on the RPN stack.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: −3 (Top Memory)
Z: 0 (3rd Level)
Y: −3 (Root 1) (1st Block)
X: −2 (Root 2) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. PLACE 1ST PIECE ON STAGE:
Type number + [ENTER] → lifts to Y

2. ADD / STACK 2ND PIECE:
Type 2nd number in X, then [+]/[×]

3. PRESS BOX ICON KEY:
Press [↱][EQN] & [↱][POLY]
✓ **Zero Parentheses • Always Visible!**

EQN BOX ICON = [↱][EQN] & [↱][POLY] • VERIFIED LESSON SEQUENCES:
- Lesson 1: 4 [ENTER] 2 [+] 4 [ENTER] 3 [+] [×] → X = 42
- Lesson 2: [↱] [POLY] 1 [A] 0 [B] 4 [+/−] [C] 0 [D] [→ X = 2

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

KEEP THIS CARD! PART IDs & VIDEO

Complete 34-Piece Reference & 60-Second 3D Video Guide

PART ID REFERENCE TABLE (34 TOTAL PIECES)

✓ KEEP THIS CARD INSIDE YOUR BOX LID!

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

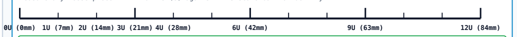
PART NAME	QTY	COLOR / FINISH	PART ID
Quadratic Tile (+x²)	1x	Green	ALGB-ALGE-27b
Quadratic Tile (−x²)	1x	Purple	ALGB-ALGE-34c
Linear Rod Tiles (+x)	5x	Green	ALGB-ALGE-2d2
Linear Rod Tiles (−x)	5x	Purple	ALGB-ALGE-2d2
Constant Units (+1)	9x	Yellow	ALGB-ALGE-76c
Constant Units (−1)	9x	Red	ALGB-ALGE-76c
Factoring Corner L-Rails	2x	Gray	ALGB-ALGE-ac0
3x1 Storage Case & Lid	2x	Black	ALGB-ALGE-49c

NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.
Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY MATCH ANY LOOSE PART

🟩 Green	🟪 Purple	🟨 Yellow	🟥 Red
Positive Tiles (+x², +x)	Negative Tiles (−x², −x)	Positive Constant (+1)	Negative Constant (−1)

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)
Measure any loose piece in 7mm units (U) against this scale bar to identify it:



✓ DISTRICT PO & SAFETY: 10-Pack SKU ALGB-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO
stackcalc.io/products/algebra-boxes/
See animated 3D examples & pack-up walkthrough

QUESTIONS, CLASSROOM SETS & SUPPORT

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

Designed in Mountain View, CA • Non-Toxic & Washable

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

BOOK CHAPTERS: SC-32 Manual Ch. 8 (Polynomial Solvers) • SC-6 Guide Ch. 8
STANDARDS: CCSS 8.EE.C.7, HSA-SSE.A.2, HSA-REI.B.4 • TEKS A.7B, A.8A (Factoring)
WATERLINE SPEC: 4-Bay Tile Deck + Dual 123mm L-Rail Frame

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SC32):
Ex 1: 4 [ENTER] 2 [+] 4 [ENTER] 3 [+] [×] → X=42
Ex 2: [↱] [POLY] 1 [A] 0 [B] 4 [+/−] [C] 0 [D] [EXE → X=2
Ex 3: [↱] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RC → X=−2

EQN **BOX ICON KEY:** [↱][EQN] & [↱][POLY]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!

◀ PAGE 07 (ECOSYSTEM) → PAGE 08 (KEEP THIS CARD FOR PART IDs & QR) ▶